



**B. Division Algorithm.** In elementary school you learned the method of long division, which can assist us for example in executing  $432 \div 15$ .

Then:  $[\text{dividend}] = [\text{quotient}] \cdot [\text{divisor}] + [\text{remainder}]$

in our case:  $432 =$

**Division Algorithm.**

Let  $a$  be an integer (the **dividend**) and  $b$  be a positive integer (the **divisor**).

Then there exist **unique** integers  $q$  (the **quotient**) and  $r$  (the **remainder**) so:

$a =$

**Proof.** Long-division provides gives us a quotient and remainder.

So, there do exist integers  $q$  and  $r$  satisfying:

$a =$

Now, we need to prove the quotient and remainder are **unique**.

To this end, let  $q'$  and  $r'$  be integers satisfying:

$a =$

To confirm uniqueness of the quotient and remainder, we must show:

We subtract our two expressions for  $a$  to obtain the following.

$0 =$

We move terms to each side and take absolute values.

Because  $r$  and  $r'$  are in  $[0 \dots b - 1]$ , it follows that:

The only multiple of  $b$  in  $[0 \dots b - 1]$  is:

Since  $|r - r'|$  has been expressed as a positive multiple of  $b$ , it follows:

Or in other words:

Uniqueness is established.

□

Technically we should provide an explanation that the method of long-division concludes in finite time, and that what it concludes with is a quotient and remainder, in the sense that  $a = bq + r$  and  $0 \leq r < b$ .

**Example 3.** Let  $n$  be an integer.

Prove that  $n^4$  is either of the form  $5k$  or  $5k + 1$ , where  $k$  denotes an integer.

**Solution.** By the division algorithm for  $n \div 5$ , we know  $n$  has form either:

$$n =$$

$$n =$$

$$n =$$

$$n =$$

$$n =$$

By the binomial theorem, we observe:

$$(5k + r)^4 = \binom{4}{0}r^4 + \binom{4}{1}(5k)^1 \cdot r^3 + \binom{4}{2}(5k)^2 \cdot r^2 + \binom{4}{3}(5k)^3 \cdot r^1 + \binom{4}{4}(5k)^4$$

and then noting that 5 may be factored out of every term but:

$$(5k + r)^4 =$$

We apply this observation to the different forms of  $n$ .

$$n^4 = (5k + 0)^4 =$$

$$n^4 = (5k + 1)^4 =$$

$$n^4 = (5k + 2)^4 =$$

$$n^4 = (5k + 3)^4 =$$

$$n^4 = (5k + 4)^4 =$$

C. **Divisibility.** We say integer  $a$  is **divisible** by pos. integer  $b$  if and only if:

- the remainder when executing  $a \div b$  is:
- or equivalently, if:  $a =$

and we denote this by:

Read this as “ $b$  divides  $a$ ”.

or alternatively say that  $b$  is a:

or that  $a$  is a:

Useful properties of divisibility are assembled in your exam **reference sheet**.

We establish a few here.

Let  $a, b, c, d$  be integers such that  $a \mid c$  and  $b \mid d$ . Then  $ab \mid cd$ .

As a special case: if  $a \mid b$ , then  $a \mid bd$ .

**Proof.** It suffices to show that  $(cd)/(ab)$  is:

We write:

$$\frac{cd}{ab} =$$

But, because  $a \mid c$  and  $b \mid d$  it follows that:

Hence:

$$\frac{cd}{ab} =$$

□

Let  $a$  and  $b$  be integers.

An **integral combination** of  $a$  and  $b$  refers to any integer of the form:

Let  $d$  be a **common divisor** of  $a$  and  $b$ .

Then,  $d$  is a divisor of any integral combination of  $a$  and  $b$ .

**Proof.** Let  $ax + by$  be an integral combination of  $a$  and  $b$ , with  $x$  and  $y$  integers.

Because  $d$  is a common divisor of  $a$  and  $b$  we may write:

$$a =$$

$$b =$$

Then:

$$ax + by =$$

is a multiple of  $d$ . Thus  $d \mid (ax + by)$ . □